PhD Lecture in Behavioral Asset Pricing: Introduction

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About Me

- Ph.D. in Finance, HKUST
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- Associate Professor of Finance
- Prior to joining CityU, taught at Singapore Management University for 8 years
- Research in behavioral finance, asset pricing, and ESG, published in top journals
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- Course materials (slides and teaching notes): github.com/franklee16/Behavioral-Asset-Pricing-Slides

Overview

- **Definition 1:**
  - Behavioral finance investigates whether some financial phenomena are the result of less than fully rational thinking
- **Three paradigms in finance:**
  - **Traditional finance**
    - Agents are rational and there are no frictions
  - **Frictional finance**
    - Agents are rational but there are frictions
  - **Behavioral finance**
    - Some agents are less than fully rational

Overview, ctd.

- **Rationality means two things:**
  - Update beliefs correctly on receipt of new information, i.e. according to Bayes' rule
  - Make decisions according to sensible preferences, i.e. according to expected utility
    - Where the utility function is defined over wealth or consumption
- **In behavioral finance, then:**
  - We study less than fully rational beliefs
    - Deviations from Bayes' rule
  - And less than fully rational preferences
    - Non-EU, or EU defined over things other than wealth or consumption

Overview, ctd. — Three Paradigms

- **Definition 2:**
  - Behavioral finance tries to understand financial phenomena using models that are *psychologically more realistic* than their predecessors
- **Three paradigms in finance:**
  - **Traditional finance**
    - Very basic, both psychologically and institutionally
  - **Frictional finance**
    - More realistic institutionally, but still psychologically simple
  - **Behavioral finance**
    - More psychologically realistic about beliefs and preferences

Challenges for Behavioral Finance

- **The arbitrage critique**
  - Claim: rational agents will quickly undo any effect that irrational agents have on prices
    - They will engage in "arbitrage"
  - We now know that there are "limits to arbitrage"
- **The lack-of-discipline critique**
  - Claim: behavioral finance is "underdisciplined"
  - It is too easy to find psychology that fits the facts
    - Address this by testing new predictions
- **Discussion:** Which critique is harder to dismiss — the arbitrage critique or the lack-of-discipline critique? Why?

Lecture Structure

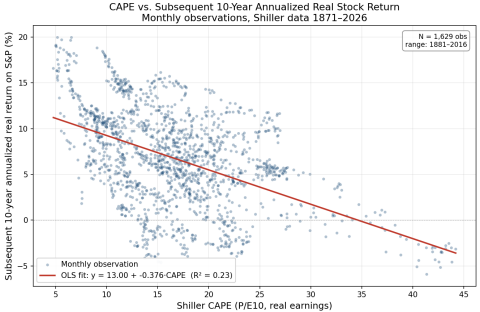
- **I. Introduction**
  - Overview
  - Facts
    - Asset price
    - Investor trading and portfolio choice
- **II. Limits to Arbitrage**
- **III. Psychological Foundation and Applications in Finance**
  - Non-standard preferences
  - Biased beliefs
  - Heterogeneous beliefs
  - Bounded rationality and psychology-free models

Roadmap

- **Asset pricing**
  - Aggregate stock market
  - Cross-section of stock returns
  - Other asset classes
  - Bubbles
- **Investor trading and portfolio choice**
  - Individual investors
  - Institutional investors

Aggregate Stock Market, ctd.

- **The time-series predictability puzzle**
  - Aggregate stock market returns are predictable in the time series
    - E.g. by the price-dividend (P/D) or price-earnings ratio (P/E)
  - This seems to contradict the efficient market hypothesis



Rational explanations?

- Rationally-varying forecasts of future risk — "rare disaster" and long-run risk models
- Time-varying risk aversion like the habit formation model (Campbell and Cochrane, 1999)

Aggregate Stock Market, ctd.

- **The excess volatility puzzle**
  - It is hard to justify stock market fluctuations on the basis of rationally-varying forecasts of the future cash flows paid to investors (Shiller, 1981)
  - A high P/D ratio does not predict higher dividend growth; instead, it predicts lower prices in the future
  - Time-series predictability and excess volatility are the same phenomenon

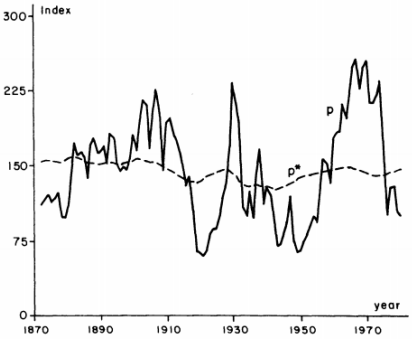


FIGURE 1

Note: Real Standard and Poor's Composite Stock Price Index (solid line p) and *ex post* rational price (dotted line p^*), 1871–1979, both detrended by dividing a long-run exponential growth factor. The variable p^* is the present value of actual subsequent real detrended dividends, subject to an assumption about the present value in 1979 of dividends thereafter. Data are from Data Set 1, Appendix.

Aggregate Stock Market, ctd.

- **The equity premium puzzle**
  - The historical equity premium (5-6%) is much higher than predicted by a simple rational, frictionless model with power utility preferences (Mehra and Prescott, 1985)
  - The covariance of stock market returns with consumption growth is small
  - The representative investor does not view the stock market as very risky, and requires a very low equity premium

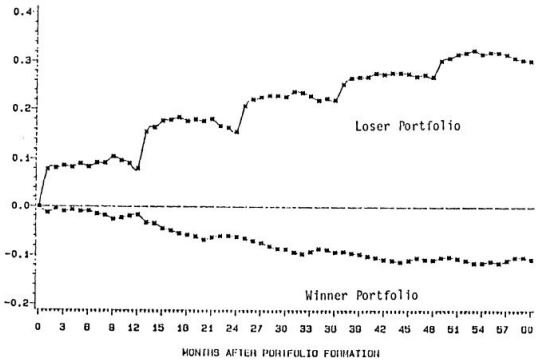
Risk premium (lognormal approximation) under consumption-based asset pricing models:

$$E_t[r_{i,t+1}] - r_{f,t} + \frac{\sigma_{i,t}^2}{2} = \gamma \operatorname{Cov}_t(r_{i,t+1},\, \Delta c_{t+1})$$

The Cross-Section of Stock Returns

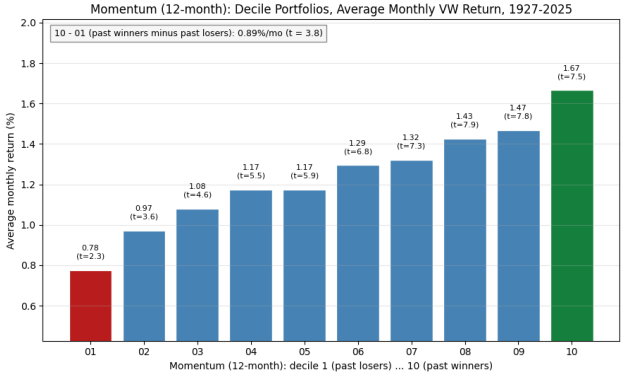
The Cross-Section — Long-term reversal

- **Long-term reversal**
  - A stock's cumulative return over the past 3–5 years negatively predicts its future return in the cross-section (De Bondt and Thaler, 1985)
  - Rational explanations?
  - What are other patterns that you notice?



The Cross-Section — Momentum

- **Momentum**
  - A stock's k-month past return (k between 3 and 12) positively predicts its future returns in the cross-section (Jegadeesh and Titman, 1993)
  - Explaining both momentum and long-term reversal in a unified and parsimonious framework is an important challenge



The Cross-Section — Short-term reversal

- **Short-term reversal**
  - A stock's return over the past week or month negatively predicts its future return in the cross-section (Jegadeesh, 1990; Lehmann, 1990)
  - Could be explained by compensation for liquidity provision
  - The returns of short-term reversal strategies in equity markets are highly predictable with the VIX index (Nagel, 2012)

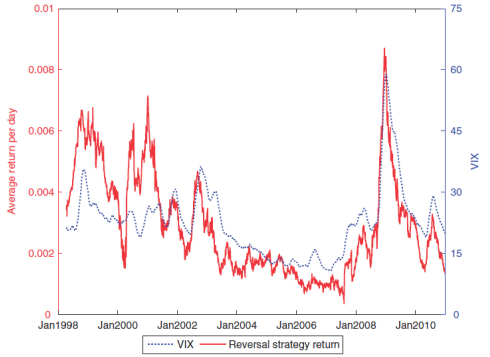
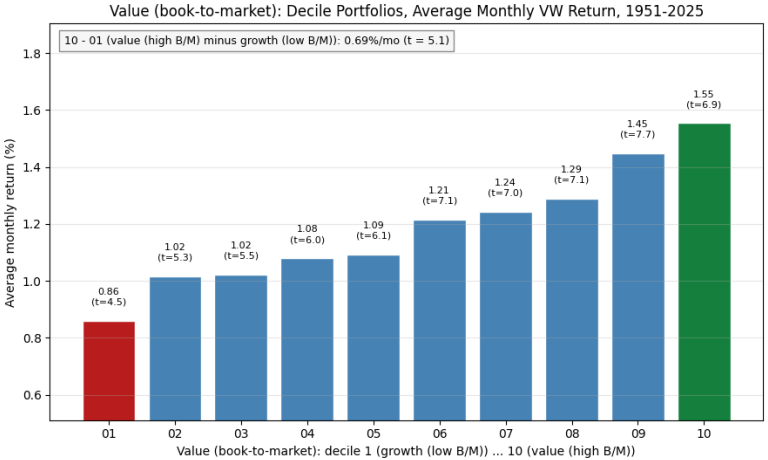


Figure 1
Three-month moving averages of daily return-reversal strategy returns and the CBOE S&P 500 implied volatility index (VIX)
Each day t, the reversal strategy returns are calculated as the average of returns from five reversal strategies that weight stocks proportional to the negative of market-adjusted returns on days $t-1$, $t-2$, ..., $t-5$, with weights scaled to add up to $1 short and $1 long. Returns are calculated from daily CRSP closing transaction prices, and returns are hedged against conditional market factor exposure.

Nagel (2012)

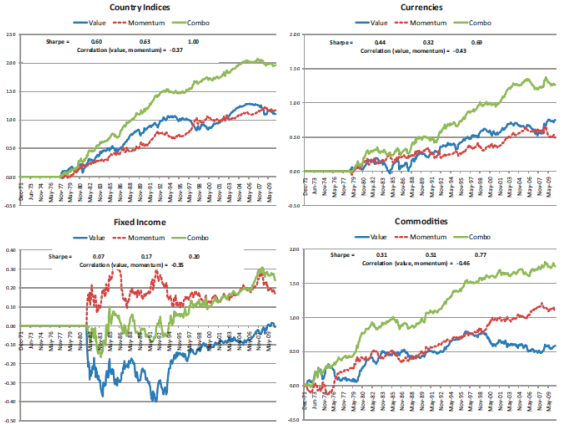
The Cross-Section — Value premium

- **The value premium**
  - Ratios of fundamentals to price (B/M, E/P, CF/P) positively predict future returns



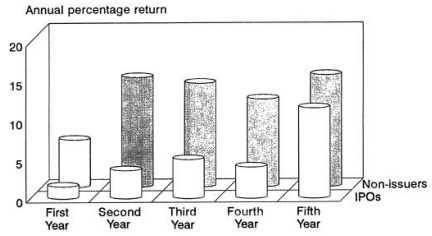
The Cross-Section — Value & momentum across asset classes

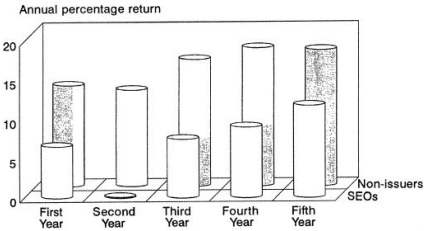
- The value and momentum effects also exist in other asset classes (Asness, Moskowitz, and Pedersen, 2013)
  - Within most developed equity markets, across country equity indices, across government bond indices, across currencies, and across commodities



The Cross-Section — Issuance/repurchase

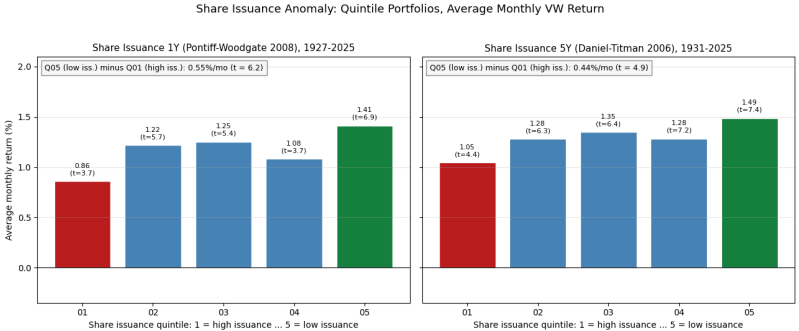
- **The issuance/repurchase anomaly**
  - Firms that issue equity (IPO or SEO) subsequently do poorly, relative to a control group (Loughran and Ritter, 1995)
  - Firms that repurchase equity subsequently do well, relative to a control group (Ikenberry, Lakonishok, and Vermaelen, 1995)





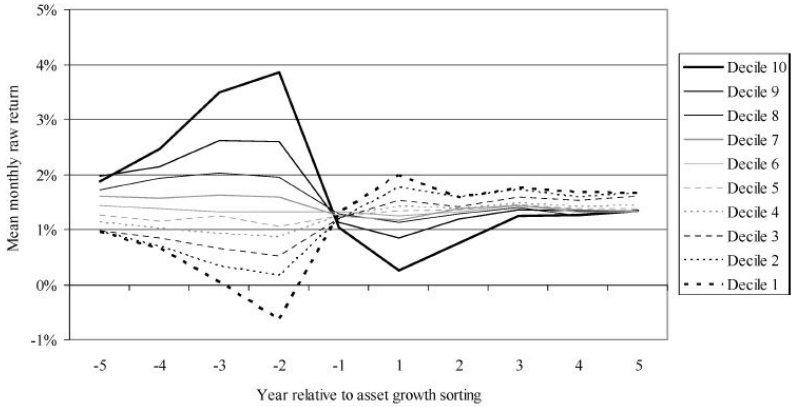
The Cross-Section — Issuance/repurchase, ctd.

- **The issuance/repurchase anomaly**
  - Daniel and Titman (2006) and Pontiff and Woodgate (2008) show net share issuance negatively predicts returns, both in the U.S. and globally



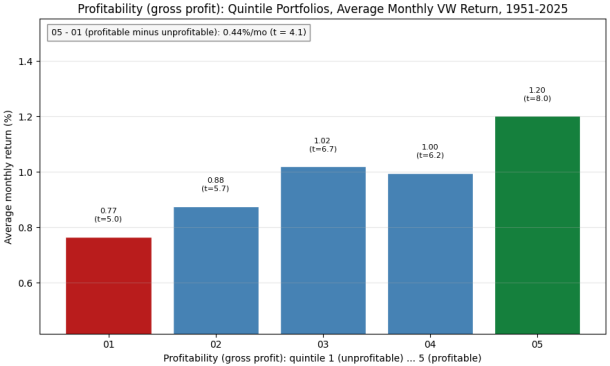
The Cross-Section — Asset growth / investment

- **The asset growth / investment anomaly**
  - Firms that grow assets more quickly or invest more have lower risk-adjusted returns (Titman, Wei, and Xie, 2004; Cooper, Gulen, and Schill, 2008)



The Cross-Section — Profitability

- **The profitability anomaly:** more profitable firms earn higher returns than less profitable firms
- Profitability is related to a broad set of *Quality* characteristics
  - Asness, Frazzini, and Pedersen (2019) suggest a multidimensional gauge of quality: profitability, recent growth in profits, low return and earnings volatility, and low leverage



- **The financial distress anomaly**
  - Financially distressed stocks earn lower alphas (Campbell, Hilscher, and Szilagyi, 2008)

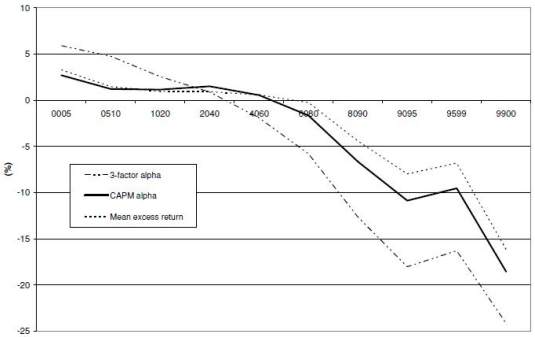
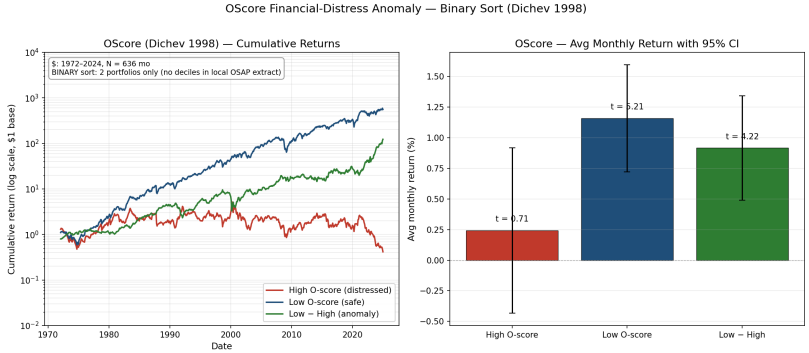


Figure 2. Alphas of distressed stock portfolios. The figure plots the annualized mean excess return over the market, CAPM alpha, and Fama–French three-factor alpha for the 10 distress risk-sorted portfolios from 1981 to 2003. Portfolios are formed at the beginning of January every year using the model-predicted probability of failure.

The Cross-Section — Financial distress, ctd.

- **The financial distress anomaly**
  - Similar findings based on the OScore measure of financial distress



The Cross-Section — Accrual and Net Operating Asset

Accrual anomaly (Sloan, 1996). Split earnings by how they are generated:

$$E_t = \underbrace{CF_t}_{\text{cash, persists}} + \underbrace{Acc_t}_{\text{accruals, mean-reverts}}$$

Investors extrapolate E_t uniformly, so they overprice high-accrual firms (receivables, inventory) and underprice low-accrual firms. As accruals reverse, prices correct: long low / short high accruals $\Rightarrow \approx 10\%$/yr abnormal return.

NOA anomaly (Hirshleifer, Hou, Teoh, and Zhang, 2004). Rank firms by the *stock* of accumulated accruals:

$$\text{NOA} = (TA - Cash) - (TL - Debt)$$

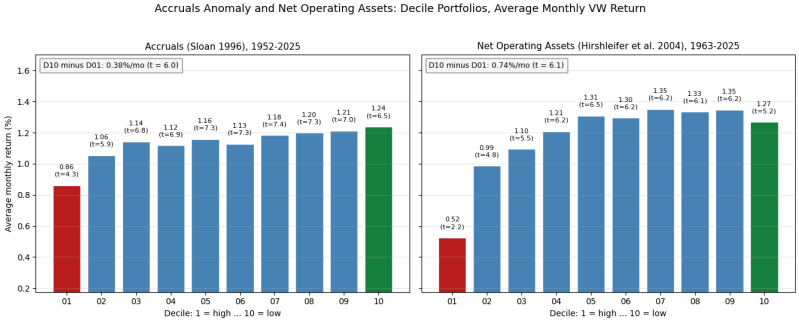
A bloated NOA is a permanent footprint of past earnings management. High-NOA firms underperform low-NOA firms by $\approx 15-17\%$/yr.

Connection: flow vs. stock.

- Sloan = *flow* (accruals, one year); NOA = *stock* (cumulative balance-sheet position) — and NOA subsumes the one-year effect.
- Single-year accruals are noisy (real expansions also build inventory); the accumulated stock isolates persistent accounting aggressiveness.

The Cross-Section — Accrual and Net Operating Asset, ctd.

- **The Accrual and Net Operating Asset Anomaly**



The Cross-Section — Post-earnings announcement drift

- **Post-earnings announcement drift**
  - The surprise in a firm's most recent earnings announcement positively predicts future returns
- **Other related findings:**
  - The revision in analysts' consensus earnings forecast/recommendations positively predicts future returns

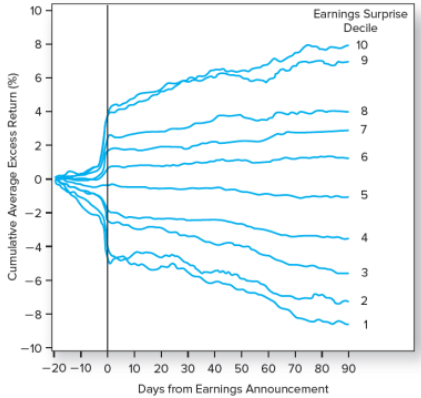
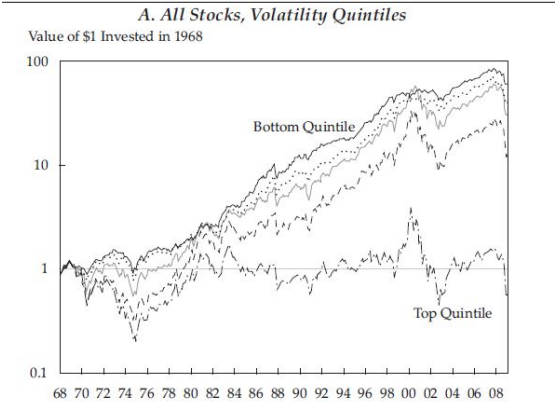


Figure 11.5 Cumulative abnormal returns in response to earnings announcements

Source: Reprinted from R. J. Rendleman Jr., C. P. Jones, and H. A. Latané, "Empirical Anomalies Based on Unexpected Earnings and the Importance of Risk Adjustments," *Journal of Financial Economics* 10 (1982), pp. 269–287.

The Cross-Section — Idiosyncratic volatility

- **Idiosyncratic volatility puzzle**
  - Stocks with higher idiosyncratic volatility have lower alphas (Ang et al., 2006)

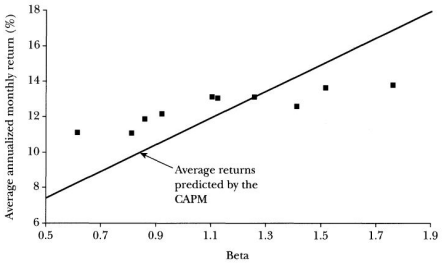


Baker, Bradley, and Wurgler (2011)

The Cross-Section — Beta anomaly

- **The beta anomaly**
  - Stocks with higher betas have similar raw returns and lower CAPM alphas than low-beta stocks
  - In other words, the empirical SML is too flat compared to that predicted by the CAPM

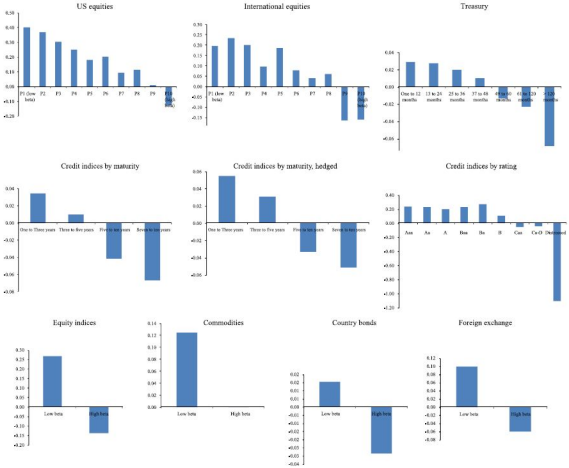
Figure 2
Average Annualized Monthly Return versus Beta for Value Weight Portfolios Formed on Prior Beta, 1928–2003



Fama and French (2004)

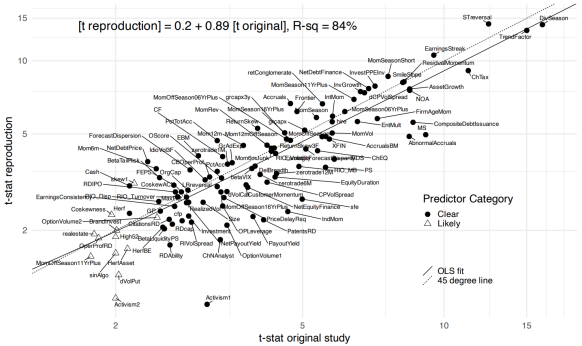
The Cross-Section — Betting against beta

- Similar findings in other countries and other asset classes (Frazzini and Pedersen, 2014)
- **Betting against beta** is highly profitable



Are Anomalies Fake?

- Some studies (Harvey, Liu, and Zhu, 2016; Hou, Xue, and Zhang, 2020) criticize the anomaly zoo, arguing most anomalies are the results of data mining and probably fake
- Still, a few influential studies (Chen and Zimmermann, 2022; Jensen et al., 2023) show that anomalies do replicate independently



Are Anomalies Fake?, ctd.

- Jensen et al. (2023) show further that most anomalies can replicate out-of-sample in other markets and the alpha magnitudes closely align with those in the U.S.

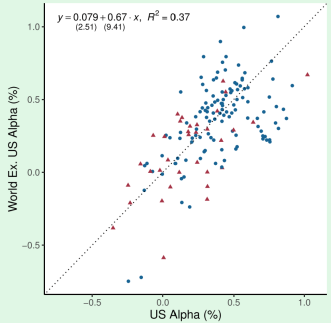
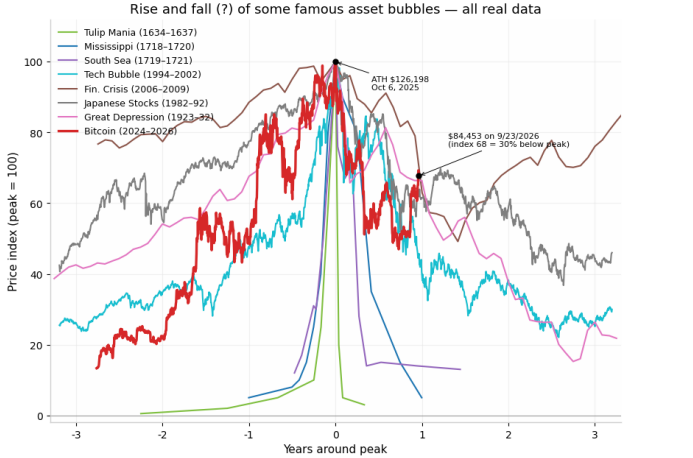


Figure 7. U.S. factor alphas versus world ex.-U.S. The figure compares OLS alphas for U.S. factors versus their international counterpart. Each world ex.-U.S. factor is a capitalization-weighted average of the factor in all other countries of our sample. Blue circles correspond to factors that were significant in the original study, while red triangles are those for which the original paper did not find a significant effect (or did not study the factor in terms of average return significance). The dotted line is the 45° line. The figure also shows a regression of world ex.-U.S. alpha on U.S. alpha. (Color figure can be viewed at wileyonlinelibrary.com)

Bubbles

- Situations where, for a significant period of time, an asset trades at a price that appears to be substantially higher than its rational/frictionless price
- Bubbles appear to exhibit the following characteristics:
  - *(i)* The price of an asset rises sharply over some period of time and then declines, reversing much of the rise
  - *(ii)* During the price rise, there are many reports in the media and from sophisticated investors that the asset is overvalued
  - *(iii)* As the price of the asset rises and reaches its peak, there is an abnormally high volume of trading in the asset
  - *(iv)* During the episode, many investors have highly extrapolative expectations about returns, in that their expectation about the asset's future return rises with the asset's past return
  - *(v)* During the price rise, some sophisticated investors increase their exposure to the asset
  - *(vi)* In the early stages of the price rise, there is positive news about the asset's future fundamentals

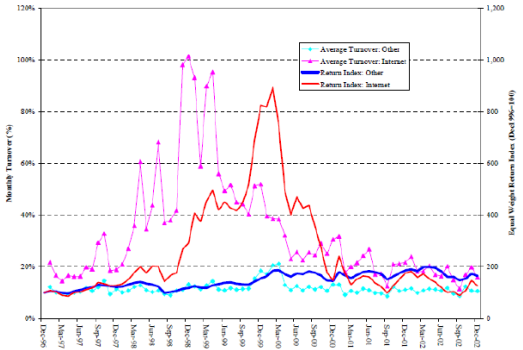
Famous Bubbles in History



Bubbles, ctd.

- Bubbles tend to be accompanied by very high trading volume (Hong and Stein, 2007)
- And sophisticated traders often "ride" the bubble (Brunnermeier and Nagel, 2004)
- Understanding bubbles is an important challenge
  - Their collapse can trigger severe economic downturns
- **Discussion:** Can rational models with time-varying risk aversion reproduce the bubble pattern?

Figure 1: Prices and Turnover for Internet and Non-Internet Stocks, 1997-2002



Investor Trading and Portfolio Choice

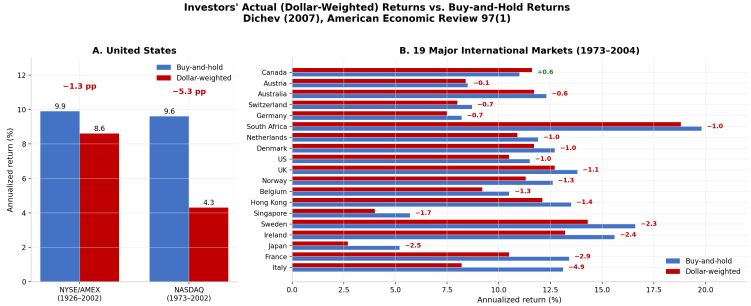
- Non-participation
- Buying high and selling low
- Under-diversification
- Excessive trading
- Sell winners too quickly and hold on to losers for too long ("disposition effect")
- Trade against anomalies

Individual Investors

- **Non-participation**
  - Historically, most U.S. households held no stock at all
    - E.g. in 1984, only 28% of households held any stock (Mankiw and Zeldes, 1991)
  - But a simple rational, frictionless model of portfolio choice predicts that all households should have at least some stock market exposure
- **Rational explanations?**
  - Fixed transaction costs of participating in the stock market
  - Correlated background risk including labor income and real estate

Individual Investors: Bad Market Timing

- **Buying high, selling low**
  - Investors' actual historical returns are much lower than the buy-and-hold returns (Dichev, 2007)



Individual Investors: Under-diversification

- Many households seem to be less diversified than recommended by traditional models of portfolio choice
- **Under-diversification: Home bias**
  - Investors allocate a surprisingly large fraction of their equity holdings to domestic equity
  - In 1989, U.S., U.K., and Japanese households invested 94%, 82%, and 98% of their equity holdings in domestic equity (French and Poterba, 1991)
- **Under-diversification: Local bias**
  - The typical U.S. household has about 30% of its portfolio invested in stocks headquartered within a 250-mile radius of the family's home (Huberman, 2001)
- **Rational explanations?**
  - Information advantage in local firms
  - Mixed evidence for this explanation (Ivkovic and Weisbenner, 2005; Seasholes and Zhu, 2010)

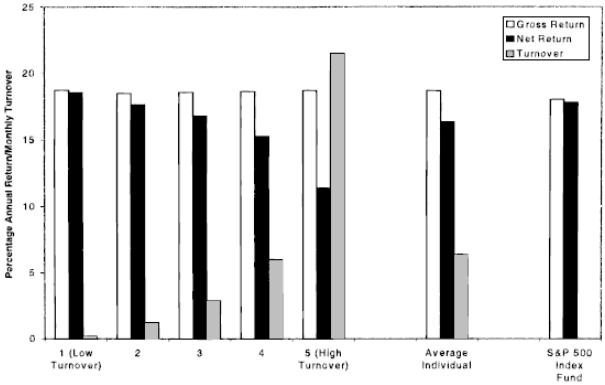
Individual Investors: Under-diversification

- **Under-diversification: Ownership of few individual stocks**
  - Curcuru et al. (2004) report that, in 1989, 30% of U.S. equity-owning households allocated more than 50% of their equity exposure to fewer than 10 stocks
- **Under-diversification: Own-company stock**
  - Benartzi (2001) reports that, in a sample of 154 retirement plans, people allocated 23% of their discretionary contributions to own-company stock



Individual Investors: Excessive trading

- **Excessive trading**
  - Using data on 66,465 households with accounts at a large discount brokerage firm, Barber and Odean (2000) find that investors who trade more earn lower returns in the stock market



Individual Investors: Disposition effect

- **Disposition effect**
  - Individual investors have a greater tendency to sell stocks trading at a gain relative to the purchase price, rather than at a loss (Odean, 1998)
- Odean (1998) tests this using the following hypothesis about investor behavior:
- Proportion of Gains Realized (PGR) $>$ Proportion of Losses Realized (PLR)

PGR and PLR for the Entire Data Set

This table compares the aggregate Proportion of Gains Realized (PGR) to the aggregate Proportion of Losses Realized (PLR), where PGR is the number of realized gains divided by the number of realized gains plus the number of paper (unrealized) gains, and PLR is the number of realized losses divided by the number of realized losses plus the number of paper (unrealized) losses. Realized gains, paper gains, losses, and paper losses are aggregated over time (1987–1993) and across all accounts in the data set. PGR and PLR are reported for the entire year, for December only, and for January through November. For the entire year there are 13,883 realized gains, 79,658 paper gains, 11,930 realized losses, and 110,348 paper losses. For December there are 866 realized gains, 7,131 paper gains, 1,555 realized losses, and 10,604 paper losses. The t-statistics test the null hypotheses that the differences in proportions are equal to zero assuming that all realized gains, paper gains, realized losses, and paper losses result from independent decisions.

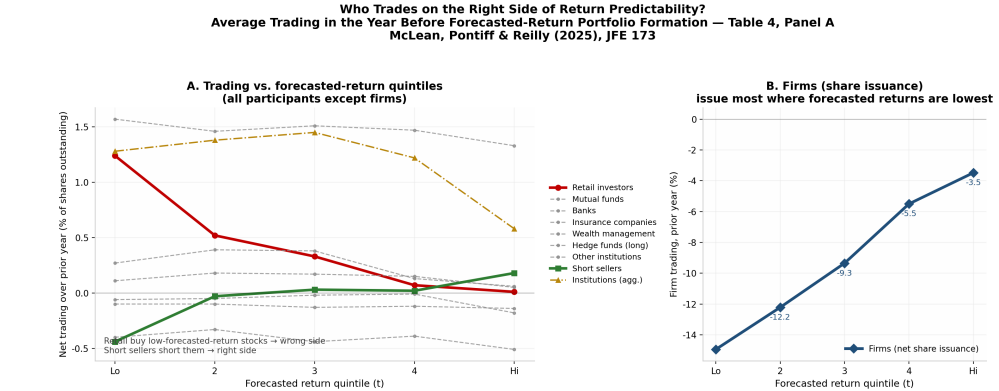
	Entire Year	December	Jan.–Nov.
PLR	0.098	0.128	0.094
PGR	0.148	0.108	0.152
Difference in proportions	−0.050	0.020	−0.058
t-statistic	−35	4.3	−38

Individual Investors: Disposition effect

- A similar effect is present among mutual fund managers (Frazzini, 2006) and also in the housing market (Genesove and Mayer, 2001)
- The most obvious potential explanations, e.g. informed trading, fail to capture important features of the data
  - Subsequent returns of winners that people sell are higher than those of losers they hold on to
- The disposition effect is more pronounced among less sophisticated investors (Dhar and Zhu, 2006)

Individual Investors: Trading aginst anomalies

- Firms and short sellers tend to be the smart money—both sell stocks with **low-forecasted returns**, retail investors systematically take the wrong side, and institutions largely trade as noise



McLean, Pontiff, and Reilly (2025)

Today and Next Time

- **Today**: a survey of empirical facts in asset pricing
  - Aggregate market: predictability, excess volatility, equity premium puzzle
  - Cross-section: significant return spread between stocks of different characteristics, which cannot be accounted for by risk exposure
  - Useful datasets to explore on your own: Open Source Cross-Sectional Asset Pricing (Chen and Zimmermann, 2022) for the U.S. market and Global Factor Returns for global markets
  - Bubbles, individual investors, institutional investors
- **Open question**: *why* do these anomalies persist?
  - Rational explanations? (time-varying risk aversion or hidden risk exposure)
  - Limits to arbitrage?
  - Behavioral mechanisms? (biased beliefs, non-standard preferences)
- *Next (Lecture 2)*: Limits to Arbitrage
  - Why rational traders may not undo mispricing — even when they see it